

# What Are We Measuring? A Benchmark Audit and a Failed Zone Hypothesis in Emotion Recognition for Autistic Children

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**Abstract**—Emotion recognition is widely proposed as assistive technology for autistic children, and papers on public autism facial-expression archives report up to 97.95% accuracy. We audit the archive behind many of those figures. It ships two overlapping dataset trees sharing 45% of the smaller tree’s images, 21.4% of its files are byte-identical duplicates, and 119 duplicate groups carry contradictory labels on pixel-identical images (103 once the unmapped *surprise* category is dropped). Holding the features and classifier fixed and changing how train and test are separated—which also moves training-set size by 2.6 $\times$  and test size by 4 $\times$ —five-category balanced accuracy spans 0.346–0.750; duplicate leakage explains little of it—a point estimate of 5.0 points, and 2.1 for a nearest-neighbour memoriser, neither distinguishable from zero at this sample size. The contradictions have a sharp geometry: one boundary, valence, absorbs 102 of 103, and a permutation null conditioned on the observed conflict count puts cross-valence conflict at a 113-fold depletion. Controls then refute our own prior three-zone framework—it absorbs none of the contradictions, because it cuts between anger and sadness where disagreement peaks, and its apparent stability advantage is generic to class count. Chance-corrected under a group-aware split, a binary valence target scores 0.780 against 0.590 for three zones and 0.625 for five categories, so this archive supports a valence decision well and finer ones poorly. This is a measurement audit, not a deployable classifier, and not a claim about any child’s internal state.

**Index Terms**—benchmark integrity, data leakage, label noise, affective computing, autism

## I. INTRODUCTION

Roughly 1 in 31 eight-year-olds in the United States is diagnosed with autism spectrum disorder (ASD) [1]. Because emotional expression in autistic people is frequently misread by non-autistic observers—the *double empathy problem* [2]—automated emotion recognition is repeatedly proposed as a communication aid.

The literature on this task is in an unusual state. Papers evaluating on public autism facial-expression archives report accuracies from 80.0% [3] to 97.95% [4]. Papers evaluating on independently collected, consented clinical corpora mostly report less: 40% for a commercial analyser on the EMBOA

autism corpus [5] and 57.95–64.92% for four commercial classifiers on parent-reported autistic children [6]. The comparison needs care. Grossard *et al.* [7] reach 82.05% within-population on lab-collected video, inside the public-archive band, and 66.43% training on typically developing children and testing on autistic ones—so part of any gap is population transfer, not data provenance. What remains unexplained is the top of the public-archive band, which no consented corpus approaches.

That correlation has become hard to ignore. Publishers have retracted work built on scraped autism face datasets, including the source paper for the archive we audit [8] and a facial emotion recognition paper [9], on the grounds that images were collected from the internet without documented diagnosis, ethical oversight, or guardian consent. That established a *provenance* failure. It did not establish whether the reported numbers were also measurement artifacts.

This paper supplies that measurement: an integrity audit of the archive as shipped (Sec. III); a protocol ablation with a nearest-neighbour reference memoriser (Sec. IV); a partition analysis locating the label contradictions, with a permutation null (Sec. V); and a control experiment that refutes our own three-zone hypothesis, leaving a narrower claim that survives it (Sec. VI). We audit an instrument rather than build one, make no claim about any child’s internal state, and treat every label as a filing decision about an image rather than ground truth about a person.

## II. RELATED WORK

**Cross-population transfer is established.** Grossard *et al.* [7] ran the controlled experiment (157 typically developing and 36 autistic children, subject-independent CV), and Gaya-Morey *et al.* [10] repeated the design with twelve CNNs on an adjacent population. We do not claim that finding; we ask whether the public benchmarks now used to measure it can support the measurement.

**Benchmark contamination is a mature field.** Kapoor and Narayanan [11] catalogue eight leakage types across 294 papers; Barz and Denzler [12] found 3.3% and 10% duplicate rates in CIFAR test sets; and Northcutt *et al.* [13] showed label errors reorder model rankings. We adopt the taxonomy of Ramos *et al.* [14]—*hard* leakage for identical images across a split, *soft* for near duplicates, *intra-dataset* for overlap within one dataset’s splits—and the notions of *augmented duplicates* [15] and *synthesis leakage* [16]. None of it examines a

TABLE I  
TWO DATASET TREES SHIPPED IN ONE ARCHIVE.

| Tree                                              | Files | Unique | Train | Test |
|---------------------------------------------------|-------|--------|-------|------|
| Balanced copy                                     | 1,425 | 1,380  | 1,199 | 226  |
| Imbalanced copy                                   | 833   | 717    | 758   | 75   |
| Shared unique images: 323 (45.0% of smaller tree) |       |        |       |      |

clinical affect dataset. Birhane and Prabhu [17] argue scraped corpora containing minors are ethically defective at collection.

**Contamination in this archive has been asserted, not measured.** The authors of the FERAC derivative [18] reported “class overlap and image duplication” here, removed duplicates, and dropped two classes as unrecoverable, but reported no detection method, threshold, duplicate rate, leakage rate, or effect of cleaning. Two groups have now flagged the same archive; what remains open is the quantity.

**Why the labels might be the problem.** Barrett *et al.* [19] challenge the premise that emotional state is readable from facial configuration at all, and Keating *et al.* [20] find autistic and non-autistic expressions differ in which muscles carry each emotion. If expressions are ambiguous under a non-autistic reading, forcing them into mutually exclusive categories should produce contradictions concentrated where categories are closest.

### III. THE ARCHIVE

We audit the public Kaggle archive associated with [8].<sup>1</sup> An earlier run of ours retained 1,672 files after excluding *surprise* and the conflicted groups. All 1,429 distinct SHA-256 digests covering those files are present in the fresh download, so the archive has not changed under us. The audit here covers all 1,774 digests. Statistics below come from the vendor’s own directory layout, before any re-splitting: this is what a downstream user receives.

**Two overlapping datasets ship as one.** The archive contains two top-level trees (Table I) with different class balances and splits, sharing 323 unique images—45.0% of the smaller tree’s unique images. Published descriptions of this dataset disagree on its size, and only one of the reported figures (830) matches either tree; two trees with overlapping contents are at least a candidate explanation, though we cannot reconstruct every published count.

**One file in five is a duplicate.** Of 2,258 labelled images only 1,774 are distinct by SHA-256; 484 files (21.4%) are byte-identical copies, in groups reaching size 5. Perceptual hashing at Hamming distance 4 adds 163 near-duplicate components. A filename suffix (*\_aug\_N*) identifies 1,020 augmentation families, 335 larger than one, so part of the archive is

<sup>1</sup>Slug fatmamtaalat/autistic-children--emotions-dr-fatma-m-talaat. We name it for reproducibility while noting it distributes images of minors of disputed provenance; readers should weigh that before downloading.

synthetic variation on fewer source images—synthesis leakage [16], and identifiable from filenames alone without any image analysis.

**Pixel-identical images carry contradictory labels.** Under the vendor’s six categories, 119 byte-identical groups (327 files) are filed under more than one emotion; dropping *surprise*, which no valence zone covers, leaves the 103 groups used from Sec. V onward. Establishing this needs no external ground truth and no annotator panel, only a hash. It is narrower than label-error detection [13]—it finds only errors a duplicate exposes, and cannot say which label is wrong—but it is assumption-free, and an internal inconsistency of the artifact is not arguable.

### IV. HOW MUCH OF THE NUMBER IS THE PROTOCOL?

Contamination matters only if it moves results. We hold the features and the classifier fixed and vary how train and test are separated. This is not a clean intervention and we do not present it as one: training pools range from 695 to 1,772 files and test sets from 69 to 293, because that is what the archive’s own splits give you. The question is what a reader learns from one published number, not what one isolated factor contributes. Features are frozen ViT-B/16 embeddings [21]; the head is a one-hidden-layer MLP (256 units, ReLU, Adam, batch 64, at most 100 epochs). Early stopping selects the epoch and the  $L_2$  penalty from  $\{10^{-5}, 10^{-4}, 10^{-3}\}$  on macro-F1 over a 15% slice *carved out of the training split* with patience 12; the test split is touched once, to score the selected model. Every protocol runs over ten seeds [22], each redrawing that validation slice. Because *surprise* has no zone, it is dropped from *both* arms, so the categorical arm is a five-way task over exactly the images the zone arm sees and the two columns differ only in label granularity.

We add a 1-nearest-neighbour classifier as a *reference memoriser*. It stores the training set exactly, so a test image whose byte-identical twin is in training is recovered with certainty. It is not an upper bound over all models—a fine-tuned network memorises and generalises, and we did not run one—but it is the estimator most able to convert an exact duplicate into a correct prediction, which makes it the sharpest available probe of how much these duplicates are worth under this representation.

Table II supports three claims and refutes a fourth—that leakage explains the published figures—which we expected to make.

**The reported number is largely a choice.** Across five defensible protocols on identical images, five-category balanced accuracy spans 0.346 to 0.750, a 40.4-point range. The protocols also differ in test-set size and composition, so this is protocol plus sampling rather than protocol alone; the smallest cells hold 69 images. Seed dispersion is 2.8–5.2 points in this block, several times smaller than the protocol spread even after allowing that the range of five draws is itself about  $2.3\sigma$ ; test-set sampling error at  $n = 69$  is roughly  $\pm 0.06$ . A reader given one number cannot tell where in that band it sits.

TABLE II

ONLY THE SPLIT CHANGES, SO TEST SIZE, TEST COMPOSITION, AND TRAIN SIZE ALL MOVE WITH IT. “EXACT/FAM.” IS THE SHARE OF TEST IMAGES WITH A BYTE-IDENTICAL TWIN IN TRAINING, THEN WITH ANY NEAR-DUPLICATE OR AUGMENTATION-FAMILY MEMBER THERE. TEN SEEDS; SD IS OVER SEEDS AND EXCLUDES TEST-SET SAMPLING ERROR ( $\approx \pm 0.06$  AT  $n = 69$ ,  $\approx \pm 0.03$  AT  $n = 293$ ).

| Protocol                                         | Bal. acc. | sd   | 1-NN  | Exact/Fam. | $n$ |
|--------------------------------------------------|-----------|------|-------|------------|-----|
| <i>Five categories</i> (chance = 0.200)          |           |      |       |            |     |
| Vendor split, imbal.                             | 0.346     | .050 | 0.348 | 1.4/26.1%  | 69  |
| Train tree A, test B                             | 0.500     | .043 | 0.476 | 23.2/24.6% | 69  |
| Vendor split, bal.                               | 0.558     | .037 | 0.585 | 0.0/33.3%  | 189 |
| Group-aware                                      | 0.700     | .052 | 0.734 | 0.0/0.0%   | 180 |
| Pooled random                                    | 0.750     | .028 | 0.755 | 35.8/80.6% | 293 |
| <i>Three zones</i> (chance = 0.333)              |           |      |       |            |     |
| Vendor split, imbal.                             | 0.497     | .063 | 0.392 | 1.4/26.1%  | 69  |
| Vendor split, bal.                               | 0.562     | .025 | 0.552 | 0.0/33.3%  | 189 |
| Train tree A, test B                             | 0.665     | .025 | 0.661 | 23.2/24.6% | 69  |
| Group-aware                                      | 0.727     | .034 | 0.733 | 0.0/0.0%   | 180 |
| Pooled random                                    | 0.735     | .023 | 0.736 | 35.8/80.6% | 293 |
| <i>Non-negative vs negative</i> (chance = 0.500) |           |      |       |            |     |
| Vendor split, imbal.                             | 0.720     | .011 | 0.569 | 1.4/26.1%  | 69  |
| Vendor split, bal.                               | 0.824     | .029 | 0.797 | 0.0/33.3%  | 189 |
| Train tree A, test B                             | 0.839     | .030 | 0.858 | 23.2/24.6% | 69  |
| Group-aware                                      | 0.890     | .019 | 0.869 | 0.0/0.0%   | 180 |
| Pooled random                                    | 0.925     | .017 | 0.926 | 35.8/80.6% | 293 |

**The zone collapse narrows the spread.** The same protocols span 23.8 points under three zones against 40.4 under five categories. Because the label spaces have different chance floors we compare them chance-corrected as  $(BA - 1/k)/(1 - 1/k)$ . Sec. VI quantifies the reduction under a single fixed split, where the controls also run, and shows it is not a property of these classes.

**Duplicate leakage is second-order at most.** Pooled random has a byte-identical twin in training for 35.8% of its test images and a family member for 80.6%, and exceeds the group-aware split by 5.0 points under five categories and 0.8 under zones. Neither difference is detectable: combining test-set standard errors at  $n = 180$  and  $n = 293$  gives  $\approx 0.047$  on the difference, so the five-category gap is about  $1.1\sigma$  and its 95% interval runs from about  $-4$  to  $+14$  points—it includes zero. We therefore cannot detect a leakage effect, and equally cannot rule out one several times larger than our point estimate. Note also that the group-aware arm holds out less (180 against 293) and so trains on more data, which if anything makes 5.0 a floor rather than a ceiling. The reference memoriser gains even less under five categories, 2.1 points. The exception is the target we go on to recommend: under binary valence the gap is 3.5 points for the head and 5.7 for 1-NN, which chance-corrected is 7.0 and 11.4—larger than the

TABLE III

DUPLICATE GROUPS STILL CARRYING CONFLICTING LABELS AFTER COARSENING, AGAINST 346 DUPLICATE GROUPS AND A 103-CONFLICT FIVE-WAY BASELINE.

| Coarsening                                | Conflicts | Absorbed   |
|-------------------------------------------|-----------|------------|
| Five categories (baseline)                | 103       | —          |
| {natural, joy} vs rest ( <b>valence</b> ) | <b>1</b>  | <b>102</b> |
| {anger, sadness} vs rest                  | 26        | 77         |
| {anger} vs rest                           | 77        | 26         |
| {anger, fear} vs rest                     | 102       | 1          |
| Valence zones (3 blocks)                  | 103       | 0          |
| Best of 15 three-block partitions         | 26        | 77         |

five-category gap of 6.2. Duplicates buy little on the hard task and more on the easy one, which is the opposite of convenient for us. 1-NN also matches or beats the trained head on four of five five-category protocols, which bounds how much the head is extracting beyond nearest-neighbour structure and is a further reason to read these absolute values as a floor.

**This form of contamination does not reach the published band.** Under five categories the best value any protocol reaches, with any estimator, is 0.755—below the 80.0–97.95% band reported on archives of this family [3], [4]. Those are plain accuracies and ours are balanced, a comparison that favours them on these imbalanced splits; our pooled-random accuracy is 0.721, which does not change the conclusion. So *exact-duplicate and augmentation-family leakage, under frozen ViT-B/16 features and our re-splits*, does not by itself produce those figures. This is not a claim that contamination is irrelevant: augmentation applied before splitting in downstream derivatives is itself synthesis leakage [16] and would not appear in our accounting, since we audit the archive as shipped. Small test sets, selection among runs, and fine-tuned networks on contaminated splits remain untested candidates.

## V. WHERE THE CONTRADICTIONS LIVE

The contradictions are not spread evenly across the label space, and locating them turns out to matter more than counting them.

Our prior three-zone framework groups the vendor’s categories as zone 1 {natural (the vendor’s label for neutral), joy}, zone 2 {anger, fear}, zone 3 {sadness}, leaving surprise unmapped. Restricting to the 1,952 files that receive a zone, we ask for each coarsening of the five categories how many byte-identical duplicate groups still straddle a block boundary. A coarsening that absorbs many contradictions is one whose boundaries the filing process respected, and since every candidate below shares a block shape, this measures *where* the boundary falls rather than how many classes there are. Table III gives the result.

**One boundary absorbs almost everything.** Splitting positive/neutral from negative leaves one conflicted group out of 346; every other binary cut leaves far more. A permutation null

preserving the zone marginals and duplicate-group structure sharpens this. The null over-predicts the total number of conflicted *files* by  $1.9\times$ , so we condition on the 280 that occur. Counting files: 113 are expected to carry a positive/neutral label against 1 observed, a 113-fold depletion resting on a single event, while files carrying a negative-affect label are *enriched* about 1.7-fold (279 against 167). (Table III counts duplicate *groups*; this paragraph counts files.) The filing process separated valence almost perfectly and anger, fear, and sadness not at all.

**The three-zone partition absorbs none of it.** This is a negative result about our own prior framework. Valence zones place anger and fear together and sadness apart, but anger versus sadness is the most common contradiction in the archive, so the zone-2/zone-3 boundary runs straight through the region of maximum disagreement. Among the 15 same-shape partitions the zone partition ties for worst at absorbing contradictions, while grouping anger with sadness absorbs 77 of 103.

We do not call this annotator disagreement: the archive is a scrape with no documented labelling procedure, and one image retrieved under two queries and written into two folders explains it as well. Either way the consequence for a model is identical, and it is the consequence our evidence supports.

## VI. WHAT THIS DOES AND DOES NOT LICENSE

**A stability argument for zones does not survive its control.** From Table II the chance-corrected protocol range narrows from 0.505 under five categories to 0.357 under three zones, a 29.3% reduction, which reads as support for zones. To test whether *these* three classes are responsible we reran the same five protocols over all 15 partitions of the five categories into blocks of 2, 2, and 1—the zones’ own block shape—on one fixed split with five seeds. There the five-category baseline is 0.482 and zones give 0.404 (16.1%; it differs from 29.3% because the control uses a fixed split and five seeds, so only within-run comparison is meaningful). But the 15 partitions span 0.325 to 0.628, a deliberately valence-crossing partition does *better* at 0.391, and zones rank 5th—above median, unremarkable, and on a statistic unlikely to be stable under reseeding. Fewer classes buys stability; these classes buy nothing extra. We report this because we expected the opposite.

**What survives is a granularity claim, and it is sharper.** The evidence in Sec. V is specific to one boundary: the reproducible distinction separates {*natural*, *joy*} from negative affect—strictly, non-negative versus negative rather than valence proper, since the vendor’s *natural* is neutral—and the anger/fear/sadness distinctions are not reproducible at all. Table II bears this out where it matters. Under the group-aware split, chance-corrected so the three label spaces are commensurable, binary valence scores 0.780 against 0.625 for five categories and 0.590 for three zones. The zone collapse does not even improve on the five-way task once the floor is removed; only the valence collapse does, and it does so at exactly the boundary the conflict analysis identifies as reliable.

So the archive supports a valence decision well and finer ones poorly, which argues for a binary target rather than the three-zone scheme we and others have used—or, if three levels are wanted for intervention, for a partition that does not cut between anger and sadness. Coarsening does reduce protocol sensitivity, but not monotonically and not in proportion to how good the target is: chance-corrected, the range is 0.505 for five categories, 0.357 for zones, and 0.410 for valence. Coarsening changes *what* is asked; it does not reliably reduce how much the answer depends on the split.

We ran no caregiver study, so the actionability that motivates zones is not something we can speak to.

**Context beyond the face.** Under our strictest split the five-category result is 0.700, and since our groups are duplicate families rather than children that is an upper bound on cleanliness. Barrett *et al.* [19] argue facial configuration underdetermines emotional state in principle. If the face reliably yields only valence, everything clinically interesting must come from elsewhere.

A face-only autism-specific model needs autism-specific face data—precisely the resource whose acquisition produced the retractions above—whereas a vision-language model supplies scene and interaction context without any such corpus. Sensing limits point the same way: microexpression analysis, the usual proposal for recovering finer states, needs frame rates and resolutions that in-the-wild capture of a child does not deliver. We report no experiment on either here; they are the direction this audit points, not results it establishes.

## VII. LIMITATIONS AND ETHICS

This audits one archive and does not establish prevalence. Duplicate detection is hash-based and misses re-encoded or cropped duplicates, so our rates are lower bounds, and the Hamming-4 threshold was not validated against hand-labelled pairs. The archive supplies no subject identities, so our groups are duplicate and augmentation families rather than people; a subject-disjoint split would be stricter. Our features come from a web-pretrained backbone, so we cannot exclude that the encoder has seen these images—a caveat our “0% leakage” rows do not escape. The reference memoriser characterises contamination under *these* features only, and no fine-tuned network was run.

**Ethics.** This is secondary analysis of an already-public dataset, with no interaction with human subjects and no re-identification attempt; we sought no institutional determination and assert none. We report only aggregate statistics, released no per-image data, and deleted the image corpus after hashing and feature extraction. We make no claim that any depicted child is autistic, that any label reflects a felt emotion, or that any system should be deployed on this basis; where Sec. V cites findings about autistic expression it motivates a hypothesis rather than asserting that this archive depicts diagnosed children. Provenance is disputed and the license unstated, so we redistribute nothing.

## VIII. CONCLUSION

This archive ships two overlapping copies sharing 45% of the smaller tree’s unique images, is 21.4% duplicated, and contains 119 groups of pixel-identical images with contradictory labels. Changing how train and test are separated moves five-category balanced accuracy across a 40.4-point band while duplicate leakage moves it by an undetectable 5.0, so one number here says little about a model.

The contradictions have a geometry: one boundary absorbs 102 of 103, a 113-fold depletion of cross-valence conflict against a marginal-preserving null, while conflict within negative affect is if anything enriched. Our own three-zone framework absorbs none of them—it cuts between anger and sadness, where disagreement peaks—and its stability advantage proves generic to class count. Chance-corrected, only the binary non-negative/negative target improves on the five-way task. This archive supports that decision and little else. Future benchmarks here should report a duplicate audit, a group-aware split, and chance-corrected metrics as a condition of being believed—and, given how this one was assembled, should not be built by scraping.

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